

Lecture #17, Nov 12, 2025

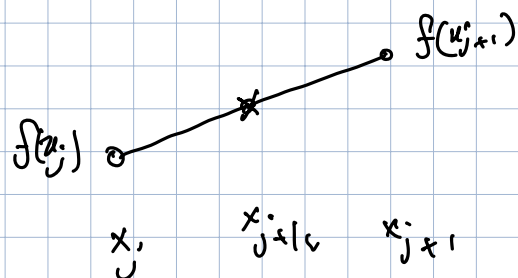
Final (exam) remark.

Flux $F_{j+1/2}^n$ at face $x_{j+1/2}$ is estimated

by u_j^n and u_{j-1}^n for 2nd order approximation.

this can be done by $u_{j+\alpha}^n$ and $u_{j-\beta}^n$, α, β integers

We use u_j^n and u_{j+1}^n by assuming a linear change.



$$\text{Linear} \Rightarrow F_{j+1/2}^n = \frac{1}{2} f(u_j) + \frac{1}{2} f(u_{j+1}) = w_1 f(u_j) + w_2 f(u_{j+1})$$

$$\text{Linear may take: } F_{j+1/2}^n = f(u_j), \quad w_1 = 1, \quad w_2 = 0$$

$$\text{Ex: } \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0, \quad a > 0$$

Derive explicit finite volume scheme assuming that

$$F_{j+1/2}^n = F_j^n, \quad \text{where } F_{j+1/2}^n = \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} f(u_{j+1/2}) dt, \quad f(u) = au$$

(Shown in lect #16 (scheme) #15 (flux approximation))

$$\text{Ans: } u_j^{n+1} = u_j^n - a \frac{\Delta t}{\Delta x} (u_j^n - u_{j-1}^n)$$

$$\text{Remark: } u_j^{n+1} = u_j^n - a \frac{\Delta t}{2 \Delta x} (u_{j+1}^n - u_{j-1}^n), \quad \text{if } F_{j+1/2}^n = \frac{1}{2} f(u_{j+1}^n) + \frac{1}{2} f(u_j^n)$$

stability, truncation error, convergence

FVM follows FDM.

FDM has limitations which could be addressed by FVM but not discussed in this course.

Ex: Lax-Wendroff scheme

$$u_j^{n+1} = \frac{1}{2} (u_{j+1}^n + u_{j-1}^n) - \frac{\Delta t a}{2 \Delta x} (u_{j+1}^n - u_{j-1}^n)$$

The modified eqⁿ is

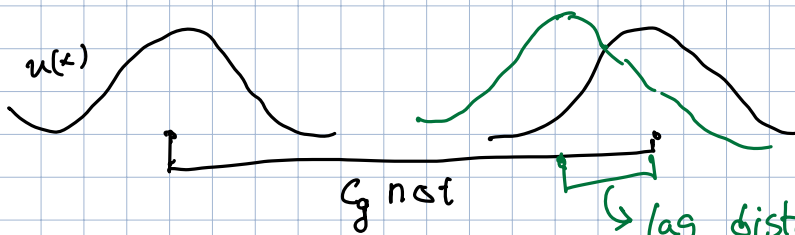
$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = -\frac{1}{6} a \Delta x^2 (1 - \tilde{C}) \frac{\partial^3 u}{\partial x^3}, \quad C = \frac{a \Delta t}{\Delta x}$$

show that the group velocity is

$$c_g = a - \frac{1}{2} a \Delta x^2 (1 - \tilde{C}) k^2, \quad u(x, t) = \int_{-\infty}^{\infty} \hat{u}(k) e^{-i(kx + \omega t)} dk$$

Ex: Estimate the group velocity and lag distance

if $k = 80$, $a = 1$, $\Delta x = \frac{1}{200}$, $C = \frac{a \Delta t}{\Delta x} = 0.8$



$$-\frac{1}{2} a \Delta x^2 (1 - \tilde{C}) k^2 n \Delta t$$

Remark

If $c_g < a$, high frequency modes will be slower than low-frequency modes.

(Example of objective type question)

Ex: Consider Lax-Friedrich method for

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

$$u_j^{n+1} = [\text{see above}]$$

show the Lax-Friedrich scheme can be written as standard form

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (F_{j+\frac{1}{2}}^n - F_{j-\frac{1}{2}}^n)$$

using

$$F_{j+\frac{1}{2}}^n \equiv F(u_{j-\frac{1}{2}}^n, u_j^n) = \frac{1}{2} [f(u_{j-\frac{1}{2}}^n) + f(u_j^n)] - \frac{\Delta x}{2\Delta t} (u_j^n - u_{j-\frac{1}{2}}^n)$$

Ex: Show that solving $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$

using Lax-Friedrich FVM is equivalent to solving

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left(v(x) \frac{\partial u}{\partial x} \right)$$

using the standard flux if $v(x) = \frac{1}{2} \frac{\Delta x^2}{\Delta t}$

Remark. Lax-Friedrich method causes artificial dissipation/damping

Von Neumann stability $u_j^{n+1} = G u_j^n$, $u_j^n = e^{ikj\Delta x}$

$$\text{Ex: } u_j^{n+1} = u_j^n - \frac{a\Delta t}{\Delta x} (u_j^n - u_{j-\frac{1}{2}}^n)$$

Von Neumann $\Rightarrow$

$$G u_j^n = u_j^n - c (u_j^n - u_{j-\frac{1}{2}}^n)$$

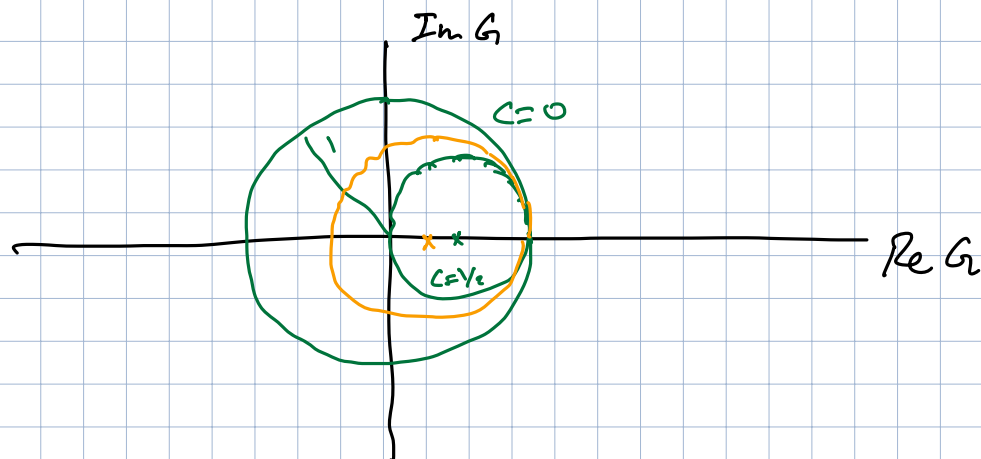
$$G e^{ikj\Delta x} = e^{ikj\Delta x} - c (e^{ikj\Delta x} - e^{ik(j-\frac{1}{2})\Delta x})$$

$$\div \text{ by } e^{iK\Delta x}$$

$$G = 1 - c(1 - e^{-iK\Delta x})$$

$$= (1-c) + c(\cos K\Delta x - i \sin K\Delta x)$$

If $c=1$, $|G|$ is unit circle



$$c=1/2, \quad G = \frac{1}{2} + \frac{1}{2}(\cos K\Delta x - i \sin K\Delta x)$$

$$\frac{(x-1/2)}{1/2} = \cos K\Delta x, \quad \frac{y}{1/2} = -\sin K\Delta x$$

$$c=3/4$$

Finite element method (FEM)

- mainly to solve elliptic problems. (i.e. steady problem)
- used in structural / stress analysis.
- very useful for "pure" numerical analysis.

FEM converts a PDE to a linear system $Au=f$

where $u \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$, $f \in \mathbb{R}^n$

Two objectives:

- 1) How to solve $Au=f$ iteratively
- 2) How to derive $Au=f$ using FEM

Iterative method for $Au = f$ (Laveque ch 4)

Defⁿ want to generate iteration matrix $G \in \mathbb{R}^{n \times n}$
and iterates u^k such that
sequence $\{u^0, u^1, \dots, u^k, u^{k+1}, \dots\}$ converges to u^*
with $Au^* = f$

if we generate sequence

$$u^{k+1} = Gu^k + c, \quad c \in \mathbb{R}^n$$

Matrix split to construct G

$$\text{let } A = M - N, \quad A \in \mathbb{R}^{n \times n}, \quad M \in \mathbb{R}^{n \times n}, \quad N \in \mathbb{R}^{n \times n}$$

$$Au = (M - N)u = f$$

$$Mu - Nu = f$$

$$Mu = Nu + f$$

$$\bar{M}Mu = \bar{M}Nu + \bar{M}f, \quad M \text{ is invertible.}$$

$$u = Gu + c$$

$$G = \bar{M}^{-1}N, \quad c = \bar{M}^{-1}f$$

Take $\boxed{u^{k+1} = Gu^k + c}$

Simplest choice is $M = D = \text{diagonal matrix.}$

$$A = \begin{pmatrix} \textcircled{D} & & \\ \textcircled{N} & \textcircled{U} & \\ & & \end{pmatrix} = L + D + U$$

$\downarrow$ Strictly lower triangular

$\rightarrow$ diagonal.

$\rightarrow$ strictly upper triangular

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{bmatrix} \quad L = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 7 & 8 & 0 \end{bmatrix}$$

Lecture #18, Nov 13, 2025

Stationary iterative methods: construction, convergence, etc.

1) stationary iterative method for solving $Au = f$

is defined as $u^{k+1} = Gu^k + c, \quad k = 0, 1, 2, \dots$

which takes the form

$$Mu^{k+1} = Nu^k + f$$

with $A = M - N$

2) If the iteration matrix G changes over the iterations then we say the method is non-stationary.

e.g. Jacobi iteration is stationary

Krylov iteration is non-stationary.

3) Error $e^k = u - u^k$ satisfies

$$e^{k+1} = G u^k$$

4) A method is convergent if $\|e^{k+1}\|$ is smaller than a tolerance ε after k -th step.

5) The process of speed up is called preconditioning

$$\|A\|_{\infty} = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|$$

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|$$

$$\|A\|_2 = \sqrt{\rho(A^T A)} \quad \text{Where } \rho(\cdot) \text{ denotes spectral radius.}$$

If λ_i be eigenvalues of A

$$\rho = \max_{1 \leq i \leq n} |\lambda_i|$$

Ex: Let $A = M - N$, show that $G = I - M^{-1}A$ is the iteration matrix.

Solⁿ

$$Au = f$$

$$Mu - Nu = f$$

$$Mu^{k+1} = Nu^k + f$$

$$u^{k+1} = \bar{M}^{-1} N u^k + \bar{M}^{-1} f$$

$$G = \bar{M}^{-1} N \Rightarrow u^{k+1} = G u^k + c$$

$$c = \bar{M}^{-1} f$$

See that $N = M - A$

$$G = \bar{M}^{-1} (M - A) = I - \bar{M}^{-1} A$$

_____ done _____

Ex: If u^k be k -th iterate

$r^k = f - Au^k$ is called residual or defect.

If $A = M - N$, show that

$$u^{k+1} = u^k + \bar{M}^{-1} r^k$$

Proof:

$$A = M - N$$

$$Au = f$$

$$Mu^{k+1} = Nu^k + f \quad (\text{could start here})$$

$$u^{k+1} = \bar{M}^{-1} Nu^k + \bar{M}^{-1} f$$

$$= \bar{M}^{-1} (M - A) u^k + \bar{M}^{-1} f$$

$$= (I - \bar{M}^{-1} A) u^k + \bar{M}^{-1} f$$

$$= u^k - \bar{M}^{-1} A u^k + \bar{M}^{-1} f$$

$$= u^k + \bar{M}^{-1} (f - A u^k)$$

$$= u^k + \bar{M}^{-1} r^k$$

Theorem : A general iterative method

$$u^{k+1} = G u^k + c$$

Converges if the spectral radius of G is less than 1 i.e. $\rho(G) < 1$

Solⁿ

$$u^{k+1} = G u^k + c, \quad G \in \mathbb{R}^{n \times n}, \quad c \in \mathbb{R}^n$$

also, $u^k = G u^{k-1} + c, \quad u^{k-1} = G u^{k-2} + c, \text{ and so on}$

Combine, $u^{k+1} = G (G u^{k-1} + c) + c$

$$= G^2 u^{k-1} + G c + c$$

$$= G^2 u^{k-1} + (I + G) c$$

$$= G^2 (G u^{k-2} + c) + (I + G) c$$

$$= G^3 u^{k-2} + (I + G + G^2) c$$

...

$$= G^{k+1} u^0 + (I + G + G^2 + \dots + G^k) c$$

let $G = \Phi \Lambda \Phi^{-1}$ Since G is non defective.

$$G G = \Phi \Lambda \Phi^{-1} \Phi \Lambda \Phi^{-1}$$

$$G^2 = \Phi \Lambda^2 \Phi^{-1}$$

thus

$$G^k = \Phi \Lambda^k \Phi^{-1}, \quad \Lambda^k = \begin{bmatrix} \lambda_1^k & & \\ & \lambda_2^k & \\ & & \ddots \\ & & & \lambda_n^k \end{bmatrix}$$

If $|x_j| < 1$ for $1 \leq j \leq n$

then $G^k \rightarrow 0$ as $k \rightarrow \infty$

then $I + G + G^2 + \dots + G^k$ converges to a limit.

Hence the method converges regardless of the choice of u^0 if $\rho(G) < 1$

We also see the $\rho(G) < 1$ is necessary and sufficient condition

Q: $G = M^{-1}N$, $A = M - N$

$$u^{k+1} = G u^k + c, \quad \rho(G) < 1$$

Can we select a G without looking at A

How many iterations needed to solve $Au = f$

Ans: We got

$$u^{k+1} = G^{k+1} u^0 + (I + G + \dots + G^k) c$$

let ρ be spectral radius of G and $\epsilon > 0$

S.t. $\rho^k = \epsilon$

$$\log \rho^k = \log \epsilon$$

$$k = \frac{\log \epsilon}{\log \rho} \text{ is the number of iteration.}$$

H.W.

$$\text{show } \|u - u^k\| \leq \varepsilon \Rightarrow K = \frac{\log \varepsilon}{\log \rho}$$

Ex:

The largest eigenvalue $\lambda = 0.9$

Tolerance $\varepsilon = 10^{-8}$, Find number of iterations

$$\begin{aligned} \text{Ans: } K &= \frac{\log \varepsilon}{\log \rho} = \frac{\log 10^{-8}}{\log 0.9} = \frac{-8 \log 10}{\log 9 - \log 10} \\ &= \frac{-8}{\log 9 - 1} = 178 \end{aligned}$$

H.W

$$u^{k+1} = G u^k + c, \quad e^{k+1} = u^{k+1} - u^*, \quad \lim_{k \rightarrow \infty} u^k = u^*$$

$$\text{show } \|e^k\|_2 \leq \rho^k K(\phi) \|e^0\|_2$$

$$\text{where } K(\phi) = \|\phi\| \|\phi^{-1}\|$$

$$G = \phi \Lambda \phi^{-1}$$

$$\rho = \max_{1 \leq j \leq n} |\lambda_j|$$